Your RAG Number Is a Property of the Template You Copied:
An Audit of the Default Question-Answering Prompts Shipped by Three RAG Frameworks

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Abstract

Retrieval-augmented generation (RAG) systems are usually built by copying a default question-answering prompt from a framework, yet the choice among those defaults is almost never reported. We take the default prompts shipped by three widely used frameworks — LangChain, LlamaIndex and FlashRAG — verbatim, and run all three on the same readers, the same documents and the same three multi-hop QA datasets. Measured exact match differs by up to **46 points** (HotpotQA: 46.4 for FlashRAG, 0.0 for the other two), while a format-insensitive measure — whether the gold answer appears anywhere in the response — differs by under five points and is not significant (McNemar $p = 0.54$). The gap is produced by response style: 19 versus 134 characters. A second, independent effect comes from one sentence inside LangChain's default, which instructs the reader to say that it does not know. With an empty context that sentence drives abstention to **79–95%**, against **0–2%** once it is deleted and nothing else changed; the deletion also removes 6–8 points of the apparent benefit of retrieval. The effect is not monotone in model size: a Llama-3.2-1B reader abstains on **91.5%** of questions *with the documents present* under that template and 0.0% under LlamaIndex's ($p = 1.6 \times 10^{-55}$). The closed-book half of our comparison is anchored against FlashRAG's published setting (HotpotQA F1 24.9 against a reported 28.4). We recommend that RAG evaluations report the template verbatim, the answer format, and abstention separately from accuracy.

Keywords: retrieval-augmented generation; evaluation methodology; prompt templates; answer extraction; abstention; reproducibility

1 Introduction

A retrieval-augmented generation system is assembled from components: a retriever, a reranker, a reader, and a prompt that glues them together [1, 4]. The first three are usually described in detail. The prompt is usually a line of code copied from a framework's default, and its exact wording is rarely reported.

This paper asks a narrow, practical question: if three practitioners build the same RAG pipeline on the same reader and the same documents, but each copies a different framework's default prompt, how far apart will their reported numbers be?

The answer, on HotpotQA [22] with a Qwen2.5-3B reader, is **46 exact-match points**: 46.4 for the FlashRAG template, and exactly 0.0 for the LangChain and LlamaIndex templates. The readers are identical, the documents identical, the questions identical.

The gap does not come from knowledge. A format-insensitive measure — does the gold string appear anywhere in the response — puts the three templates within five points of each other, with no significant difference between the two extremes. The models produce the same content; one template asks for a short span and gets one, the other two ask for an explanation and get 134 characters instead of 19. Under exact match, the second kind of response scores zero even when the answer is inside it.

A second effect runs in the opposite direction and is not a measurement artefact at all. LangChain's default contains the sentence *"If you don't know the answer, just say that you don't know, don't try to make up an answer."* With an empty context, that sentence drives the reader to refuse on 79–95% of questions. Deleting that sentence and nothing else drops abstention to 0–2%, while the format-insensitive measure barely moves: the sentence changes whether the reader speaks, not what it would have said. The practical consequence is that the "closed-book baseline" reported in many RAG papers measures a refusal policy installed by the template, and that it inflates the measured benefit of retrieval by six to eight points in our data.

The two effects belong to different classes. The first is an evaluation artefact of the kind analysed by Sclar et al. [7], Voronov et al. [8], Mizrahi et al. [9] and Hua et al. [11]: the model knows the answer, the metric cannot see it. The second is a behavioural change: the model abstains because it was told to. Reporting the number without distinguishing them is what makes RAG results hard to compare.

Contributions

1. We take three published default templates verbatim, with provenance, and measure what a practitioner copying each would report: a 46-point exact-match spread on identical inputs (Section 5.1).
2. We show the spread is response style rather than knowledge, using a containment measure and a truncation probe that rules out the "short answer buried in prose" explanation (Section 5.2).
3. We isolate one sentence of one template with a byte-level ablation and three control variants, and show it controls the closed-book baseline (Section 5.3).
4. We characterise reader dependence across two model families and four sizes, and find that compliance is a property of the family as much as of the size (Section 5.4).
5. We anchor the closed-book half of the comparison against a published evaluation setting (Section 5.5).

We claim no new method. Every component is standard: BM25 retrieval, greedy decoding, exact match and token F1. The contribution is an audit of defaults, and it stands or falls on whether the measured spread reproduces — which is why every number in this paper is regenerated from raw prediction files by a script.

2 Related work

Prompt and format sensitivity. Sclar et al. [7] show that spurious features of prompt formatting can move benchmark accuracy by tens of points, and introduce a spread metric to quantify it. Voronov et al. [8] make the same point for in-context learning comparisons, and Mizrahi et al. [9] argue for multi-prompt evaluation as a default. Webson and Pavlick [10] examine whether models actually use prompt meaning or surface form. Our Section 5.1 is a concrete instance of this literature, measured on a different object: not templates the authors wrote, but the three defaults that frameworks ship, which is what a practitioner actually copies.

Evaluation as an artefact source. Hua et al. [11] show that much of the reported prompt sensitivity originates in heuristic evaluation, in particular rigid answer matching, and that it shrinks under LLM-as-a-judge [12] scoring. Our result is the same mechanism with a measured magnitude on shipped defaults. Broader work on reproducible evaluation [13, 14] makes the complementary point that harness-level choices dominate many reported differences.

Table 1: The three default QA templates, verbatim, with provenance.

id	source	abstention clause	output cue
LC	`langchain.chains retrieval_qa.prompt`	**yes**	`Helpful Answer:`
LI	`llama_index DEFAULT_TEXT_QA_PROMPT`	no	`Answer:`
FR	FlashRAG `base_prompt` (answer prompt)	no	`Answer:`

Abstention and calibration. A line of work documents that a model's willingness to abstain is largely installed by the prompt rather than by its knowledge [15], and that models can estimate their own uncertainty when asked in the right way [16, 17, 18]. We extend the instructional account from the retrieval condition to the *closed-book baseline*: the template most practitioners copy makes that baseline near-silent.

RAG systems and toolkits. Retrieval-augmented generation spans dense retrieval [2], fusion-in-decoder readers [3], and adaptive or self-reflective variants [5, 6]. FlashRAG [19] documents its evaluation setting explicitly — Llama3-8B-instruct, e5-base-v2, five documents — and reports HotpotQA F1 of 35.3 for standard RAG and 28.4 without retrieval; we use those as the anchor in Section 5.5. LangChain [20] and LlamaIndex [21] ship the templates of Section 3 as their default retrieval-QA prompts. None of these projects is at fault for the numbers; the point is that the choice among them is unmade and unstated in most downstream work.

Benchmarks. We use HotpotQA (distractor setting) [22], 2WikiMultiHopQA [23], and MuSiQue [24], with exact match and token F1, the metrics established by extractive QA benchmarks [25], and follow the multi-hop evaluation conventions of Bamboogle [26].

3 The three published defaults

We did not write a prompt. Table 1 lists the three templates, reproduced verbatim from their sources, with the sentence that matters and the output cue that shapes the response.

The LangChain template reads, verbatim:

> Use the following pieces of context to answer the question at the end. If you don't know the answer, just say that you don't know, don't try to make up an answer.
> {context}
> Question: {question}
> Helpful Answer:

The abstention sentence is the second sentence; the output cue is the last line. Both will matter, for different reasons.

4 Experimental setup

Readers. Four instruction-tuned models served locally through a quantised runtime at 4 bits: Llama-3.2-1B, Qwen2.5-3B (primary), Qwen2.5-7B, and Llama3-8B. Decoding is greedy with a fixed seed and a generation cap of 32 tokens. The model families are those of Qwen [27] and Llama 3 [28].

Table 2: Retrieval condition, Qwen2.5-3B, $n = 500$ per cell. Containment is within five points across templates while EM spans 12–46 points.

dataset	template	EM	F1	containment	abstention
HotpotQA	FR	**46.4**	58.2	50.6	0.0
	LI	0.0	12.3	47.8	0.0
	LC	0.0	13.3	52.2	0.8
2Wiki	FR	**37.8**	44.4	44.4	0.2
	LI	0.0	10.2	40.4	0.2
	LC	0.0	11.8	43.0	6.2
MuSiQue	FR	**12.2**	19.9	15.2	1.0
	LI	0.0	4.5	10.4	0.0
	LC	0.0	4.5	10.2	6.0

Data. HotpotQA (distractor), 2WikiMultiHopQA, and MuSiQue held-out slices, 500 questions each, with a frozen question list. Each question is accompanied by its ten candidate paragraphs; we order them by BM25 and present all ten.

Conditions. *rag*: the ten documents are inserted into the template. *closed*: the context slot is empty and nothing else changes.

Clause ablation. Four variants of LC differing only in that sentence: `LC` unchanged (209 characters), `LC_minus` sentence deleted (117), `LC_cleared` replaced by the empty string (118), and `LC_neutral` replaced by an equal-length neutral sentence (217). Four variants, one variable.

Metrics. Exact match (EM) after standard normalisation; token F1; *containment* (does the gold answer appear anywhere in the response, case-insensitive); and *abstention* (the response is empty or is a refusal). Containment is reported because those templates elicit prose, for which EM is uninformative by construction.

Scale. 21,000 reader calls in total. All predictions are stored per question and every number in the paper is regenerated from them by a single verification script, which recomputes 40 quoted values and compares them with the manuscript.

5 Results

5.1 The template decides whether EM is measurable

Table 2 reports the retrieval condition with the Qwen2.5-3B reader.

On HotpotQA the FlashRAG template reaches 46.4 EM and the other two exactly 0.0, a difference of 46 points on identical inputs. Containment, however, is 50.6, 47.8 and 52.2; the paired difference between FlashRAG and LangChain is -1.6 with an exact McNemar $p = 0.54$, and between LlamaIndex and LangChain -4.4 with $p = 0.008$.

The same pattern holds on the other two datasets: EM spreads by 12–38 points while containment stays within five. Figure 1 repeats the comparison across all four readers; in every case the FlashRAG template is far above the other two.

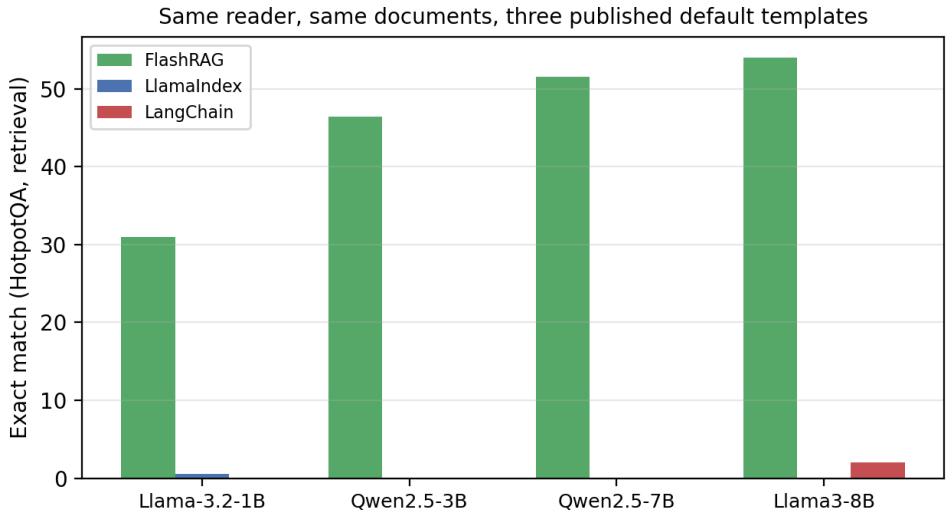


Figure 1: Exact match on HotpotQA (retrieval condition) for the same reader, the same documents and the same questions, under three published default templates.

Table 3: Response length, HotpotQA, retrieval condition. The seven-fold difference in length is the whole of the EM gap.

template	mean chars	median chars	EM	containment
FR	**19**	12	46.4	50.6
LI	134	136	0.0	47.8
LC	133	134	0.0	52.2
LC_minus	139	140	0.0	48.4
LC_neutral	141	142	0.0	45.4

A practitioner who copies LangChain's default would conclude, from any of these three datasets, that the reader cannot answer the question at all.

5.2 Mechanism: response style, not knowledge

Table 3 reports mean response length. The FlashRAG template yields 19-character answers; the other two yield 133–141.

One might assume that a short answer is buried at the front of the prose and simply needs to be extracted. It is not. Truncating every response to its first w words and re-scoring gives Table 4.

Truncation recovers at most 1.4 points. The correct string is present somewhere in about half of the LangChain responses (containment 52.2), but the model never leads with it: the template shifts the task from extraction to commentary. Under a strict-match metric, a commentary is worth zero.

5.3 The clause decides what the closed-book baseline measures

LangChain's default contains one sentence that no other template in our set contains. With an empty context, that sentence dominates the behaviour (Table 5).

Deleting that one sentence — leaving the template byte-identical otherwise — moves abstention from

Table 4: Exact match after truncating the response to its first w words.

template	full	$w = 3$	$w = 5$	$w = 8$	$w = 15$
FR	46.4	41.8	45.8	46.6	46.6
LI	0.0	1.4	0.2	0.0	0.0
LC	0.0	1.2	0.2	0.0	0.0
LC_minus	0.0	0.8	0.2	0.0	0.0

Table 5: Abstention rate with an empty context, Qwen2.5-3B, $n = 500$. The clause is the whole difference.

dataset	LC	LC_minus	LC_neutral	LC_cleared
HotpotQA	**91.2**	1.0	0.2	1.4
2Wiki	**79.0**	0.0	0.2	0.0
MuSiQue	**95.0**	2.0	0.0	1.6

79–95% to 0–2%. The equal-length neutral replacement reproduces the collapse, so the effect is the sentence's content and not its length.

Critically, containment in the same condition barely moves: 11.2 under LC and 14.4 under LC_minus on HotpotQA (28.6 and 31.8 on 2Wiki). The sentence does not change what the reader would say; it changes whether the reader says anything.

The consequence for a standard comparison is in Table 6: the clause inflates the apparent benefit of retrieval by 8.2 points on HotpotQA and 6.0 on 2Wiki.

5.4 Reader dependence

We repeated the comparison for four readers. Table 7 shows that the two effects behave differently.

The format effect is universal. At every reader, the prose-eliciting templates collapse to zero or near-zero EM while FlashRAG's reaches 31–54.

Compliance varies by family as much as by size. Qwen2.5-7B abstains on 0.0% of questions with documents present; Llama3-8B, on the same template and data, abstains on 22.0%. A 7B model from one family resists an instruction that an 8B model from another obeys.

The smallest reader is the most compliant. Llama-3.2-1B answers 91.5% of the time with *no* documents, and refuses 91.5% of the time when the documents are right there under LangChain's template. The paired comparison against LlamaIndex's template is decisive: 91.5% versus 0.0% abstention, exact McNemar $p = 1.6 \times 10^{-55}$ with 183 discordant pairs, all in one direction, and containment 6.5 versus 44.5.

5.5 Anchor against a published setting

An audit is only useful if its instrument is calibrated. FlashRAG publishes its reader configuration — Llama3-8B-instruct, e5-base-v2, five documents — together with HotpotQA F1 of 35.3 for standard RAG and 28.4 without retrieval. Table 8 compares our run of that reader under that template.

The closed-book condition agrees to within 3.5 points, and it is the condition in which there is no retriever, hence nothing else that could differ. That is our calibration point: the reader, the prompt and the metric behave like the published pipeline.

The retrieval condition does not agree, and the reason is known: their retriever scores five paragraphs from a one-million-document Wikipedia corpus, whereas our pool contains the gold paragraph by construction. In a separate experiment in our project, open-corpus retrieval placed the full gold paragraph set in the

Table 6: Retrieval minus closed book, in containment points. The clause inflates the apparent benefit of retrieval.

dataset	LC	LC_minus	LC_neutral	LC_cleared
HotpotQA	41.0	34.0	34.2	32.8
2Wiki	14.4	8.4	8.2	11.8
MuSiQue	10.2	9.2	8.6	11.0

Table 7: HotpotQA, retrieval condition unless stated, $n = 200$ per cell.

reader	template	EM	containment	abstention
Llama3-8B	FR	**54.0**	56.5	0.0
	LC	2.0	45.5	**22.0**
	LI	0.0	42.5	0.0
	LC (closed)	0.0	6.0	**80.0**
Qwen2.5-7B	FR	51.5	57.5	0.0
	LC	0.0	52.5	0.0
	LC (closed)	0.0	17.0	15.0
Qwen2.5-3B	FR	46.4	50.6	0.0
	LC	0.0	52.2	0.8
	LC (closed)	0.0	11.2	**91.2**
Llama-3.2-1B	FR	31.0	33.0	0.0
	LC	0.0	**6.5**	**91.5**
	LI	0.5	44.5	0.0

top ten for only 59% of questions, against 100% for the pool. That gap, not the reader, accounts for the difference.

6 Discussion

Taken together, the two effects say something simple: *the reported number is a property of the template, and the template is rarely reported.*

The format effect is an evaluation artefact in the sense of Hua et al. [11]: the knowledge is there, the metric cannot see it. It is nevertheless an artefact that produces a 46-point difference between two systems that are, in every other respect, identical, and it is not hypothetical — it is the difference between the defaults that two of the most widely used frameworks ship.

The clause effect is not an evaluation artefact at all: removing it changes behaviour, not scoring. It matters because the closed-book condition is the standard evidence that "retrieval helps", and in the default configuration that condition is largely measuring a refusal policy. The inflation we measure, 6–8 points, is of the same order as the differences RAG papers routinely claim.

Recommendations.

1. Report the template verbatim. "We use a standard RAG prompt" spans a 46-point range.
2. Report abstention separately from accuracy. The identity $\mathrm{EM} = (1 - u)\, c$, with u the abstention rate and c the conditional accuracy, makes the policy term explicit.

Table 8: Anchor. The closed-book condition, which has no retriever, agrees to within 3.5 F1.

condition	retriever	ours (F1)	FlashRAG reported (F1)
closed book	none	**24.9**	**28.4**
retrieval	ours: 10-doc pool	67.6	35.3 (e5, 1M corpus)

3. If exact match is the metric, use a short-answer cue; otherwise report F1 or containment, because prose responses score zero by construction.

4. When claiming an instruction effect, include a neutral-sentence control: our equal-length replacement reproduces the abstention collapse, which rules out length.

7 Threats to validity

Scale. Two model families and four sizes, all quantised to 4 bits, on one machine. The absolute levels are specific to these readers.

Retriever. One sparse retriever over a per-question pool. The retrieval condition is therefore not comparable with open-corpus results, as Section 5.5 shows explicitly. All our claims are within-stack comparisons.

Templates. Three templates is a sample, not a census; other frameworks may ship other defaults. The clause ablation exists only for LangChain, because the other two templates have no abstention sentence to delete — which is itself part of the finding.

Generality. We test English multi-hop QA. Whether the same spread appears in single-hop QA, in other languages, or with API-based readers is untested.

8 Conclusion

Three frameworks ship three different default prompts, and on the same reader and the same documents those prompts produce exact matches of 46.4, 0.0 and 0.0, while a format-insensitive measure puts them within five points of one another. One sentence inside one of those defaults additionally drives the closed-book baseline to near silence, and inflates the apparent benefit of retrieval by six to eight points. Neither effect is exotic; both follow from defaults that are copied without being read. Reporting the template, the answer format and the abstention rate costs three lines and makes the number interpretable.

Declarations

Funding. Not applicable.
Competing interests. The authors declare no competing interests.
Ethics approval. Not applicable; the study uses public benchmarks and locally served open-weight models.
Consent to participate / for publication. Not applicable.
Data availability. All datasets are public (HotpotQA [22], 2WikiMultiHopQA [23], MuSiQue [24]). Per-question predictions and the scripts that regenerate every table are available in the project repository; see Code availability.
Code availability. The runner that produces the predictions, the analysis scripts, the figure scripts, and the verification script that recomputes all 40 quoted values are released at https://github.com/910244720/rag-template-audit (to be made public upon acceptance).

Author contributions. Fangqi Liu: conceptualisation, methodology, software, investigation, formal analysis, writing — original draft. Yuhe Liu: data curation, validation, writing — review and editing. Both authors read and approved the final manuscript.

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